



Chapter 2:

The Digital Forensics Process



Outlines

- Introduction
- Why do we need a process?
- Principles of a forensics process
- The phases (identification, collection, Examination, Analysis, and Presentation)
- Summary.



...Refresher

- Crimes can happen in various settings—work, school, or social situations—can now involve digital devices and services.
- Many principles from physical investigations and forensic science apply equally well in digital forensics.
- Digital evidence may be found in **computers, mobile devices, internet infrastructure, industrial systems, and other digital devices**.
- We see the five phases of the digital forensics investigation process, grounded in **digital forensics principles and standard law enforcement and industry practices**.
- While technology changes, motivations for crime remain consistent, with traditional crimes often having digital parallels (e.g., hacking an online store versus robbing a physical one). Digital services may also highly enable criminal planning or communication.



Introduction

- Cybercrime has evolved significantly over the past decade, with attackers becoming more technically savvy and better incentivized, using sophisticated methods to exploit networks and endpoints.
- Data breaches resulting from cybercrime highlight the need for **clear forensic investigation processes**, along with **appropriate tools** to assess incident causes and effects.
- Established forensic principles for cybercrime investigations guide areas like **computer forensics**, **internet investigations**, and **mobile device forensics**, supported by computational tools and legal compliance.
- Trusting information and evidence in digital forensics requires **addressing uncertainties** in potential evidence, both physical and logical, accidental and deliberate.



Example: Email from Whom?

- Imagine investigating a case where a person has received an email with potentially important evidence of a crime.
- The email was sent over the Internet. Because of this, the origin of the email must be considered uncertain, as must the timestamps, since they could have been tampered with while en route.
- If the email was in fact sent from a given system, how may one know that it was created by the system's owner and not by some intruder intentionally placing it there?
- What if the email was sent by a Trojan horse or other malware? This argument can be used as part of a legal defense, often referred to as the Trojan horse defense (Brenner et al., 2004).



Why Do We Need a Process?

- The forensic process involves a structured investigation of digital evidence from any device that stores or processes data in digital form, sharing similarities with physical investigation processes.
- A key difference in digital forensics is that the evidence is digital, requiring specific approaches to gather and analyze it.
- The challenge is *how to* gather and *what* to gather as the evidence will answer the **5WH** questions.
- Proper management of identified digital evidence is essential to support a hypothesis of a crime or incident and to prove a case.



Principles of a Forensics Process

- The investigation should **focus on identifying the strongest evidence possible with available resources while documenting the processes, key assumptions, and uncertainties.**
- **Numerous tools exist** for collecting, examining, analyzing, and presenting digital evidence, developed by commercial entities, the digital forensics community, and academic researchers.
- **Information Network Security Agency (INSA)** is the primary body for cybercrime investigation and building "digital trust" in Ethiopia.
- But The **Ethiopian Standards Institute** references **international standards** (such as ISO/IEC 27037) which govern the identification, collection, and preservation of digital evidence to ensure integrity and authenticity. (ISO/IEC 27000 family of standards focused on DS)
- Key forensic principles include evidence integrity, which involves preserving evidence in its original form, and chain of custody, which documents all actions taken with the evidence to ensure its authenticity.



Finding the Digital Evidence

- Any digital data that contains reliable information ***supporting or refuting a hypothesis*** about an incident or crime can be considered as evidence.
- **Electronic Evidence:** A related term, electronic evidence, refers to “electronically stored information admitted as evidence at a trial or hearing”
- **Forensic analysts** process digital objects to determine their relevance to an investigation
- Digital evidence is recognized as relevant only after it has been determined through analysis, in line with the ISO27042 standard; prior to this, it is merely potential evidence.
- Digital evidence can be relevant in two ways: **ontologically**, as something observable and describable, and as **evidence** in the way of recognition, which reveals insights about a case.



Finding the Digital Evidence

- According to the ISO27042 standard, digital objects are only considered "potential evidence" until they are analyzed to determine their relevance:
 1. **Ontological Relevance:** This refers to the evidence as a measurable, physical, or logical object. It is about describing the data's existence without necessarily knowing what it proves yet.
 2. **Relevance by Recognition:** This is where the evidence starts to tell a story or "reveal insights about a case". This happens during the Analysis phase, where you determine the facts of an event and who might be responsible.

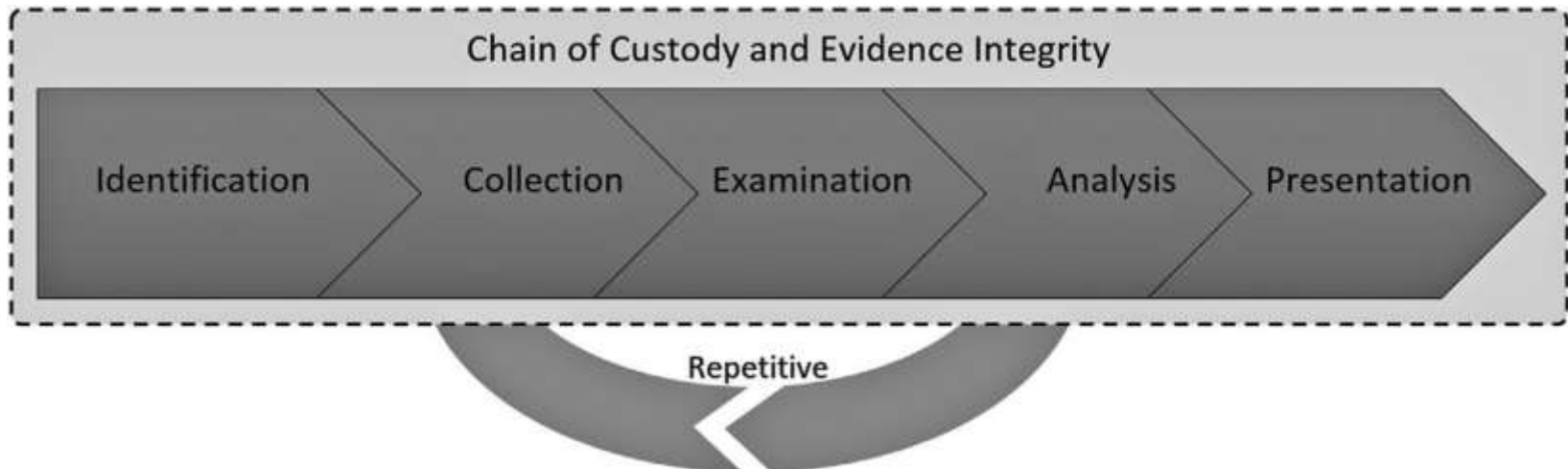
Example: You find a file named ledger.xlsx on a hard drive.

- **The Ontological View:** observe and describe its properties: it is a 50KB spreadsheet file, created on January 10th, located in the "Documents" folder.
- **The Recognition View:** (open that ledger.xlsx file) and see what the spreadsheet contains a list of illegal transactions and the suspect's login credentials. This is to support a hypothesis and help answer the 5WH questions (Who, What, When, Where, Why, How).



Introducing the Digital Forensics Process

- The digital forensic process is based on a structured, *normative approach*, similar to traditional physical forensic investigations.
- **Iterative Phases:** The process involves five key phases—**identification, collection, examination, analysis, and presentation**—allowing for multiple iterations to refine findings.





Introducing the Digital Forensics Process...

The digital forensics process typically starts with an observation, alert, or notification, leading to a hypothesis and the beginning of the investigation.

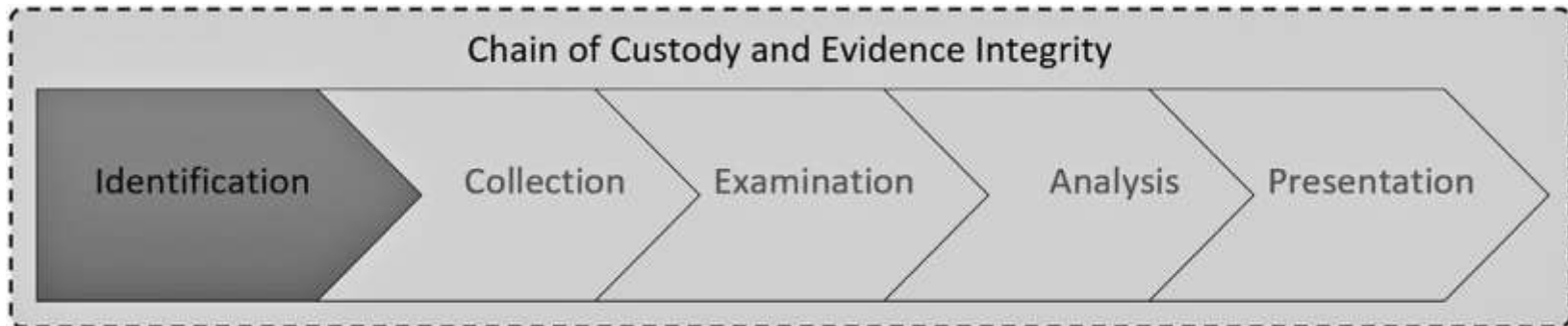
Standardization Efforts: Since the 1980s, research focused on standardizing digital forensic investigation processes, identifying common characteristics and phases, with this book outlining the five most common phases.

- Adhere to the leading practice and theory from traditional forensic investigations.
- Be easily adaptable and practically oriented, and support traditional investigation steps, and Support forensic reconstructions
- Remain independent of technology, product, and procedure.
- Applicable to law enforcement, corporate, and incident investigations.
- Improve proper and efficient identification of facts.
- Limit unnecessary generation of new artifacts.
- Use and support forensic science and technology, and the results must be usable in a court of law.
- Must be capable of being scrutinized by the judiciary if the case is presented in court.



The Identification Phase

- It is the task of **detecting, recognizing, and determining the incident or crime to investigate.**
- Incidents can be identified based on complaints, alerts, or other indications.
- It lays the groundwork for all subsequent investigation activities by *pinpointing relevant evidence and objects to examine.*
- **Then,** form a hypothesis about the incident, focusing on evidence that may support, refute, or validate the case.
- **Finally,** Investigators use the 5WH model (Who, What, When, Where, Why, How) to guide hypothesis development, especially when determining the status of files on digital devices or systems.

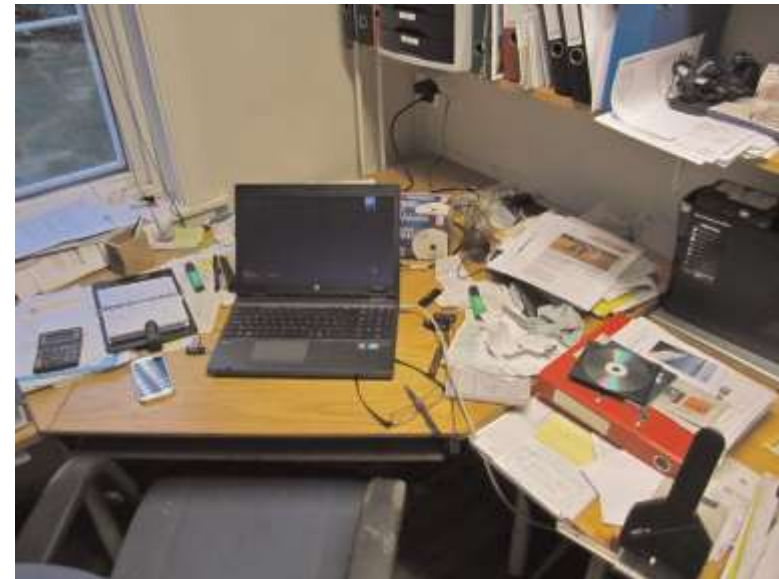




The Identification Phase...

An investigation can aim to find information that supports, refutes, or verifies a case.” Here are some important activities in identification.

- **Preparation:** Plan and deploy tools ahead of time to ensure investigation quality.
- **First Responders:** Act with precision at the scene to prevent misleading the inquiry.
- **Scene Assessment:** Analyze the environment thoroughly to understand the event context.
- **System Status:** Categorize equipment as live (turned on devices) or dead (off) to determine the capture method.
- **Chain of Custody:** Maintain rigorous documentation to track evidence handling and history.





The Collection Phase

Collection of data from digital devices to **make a digital copy using forensically sound methods and techniques.**

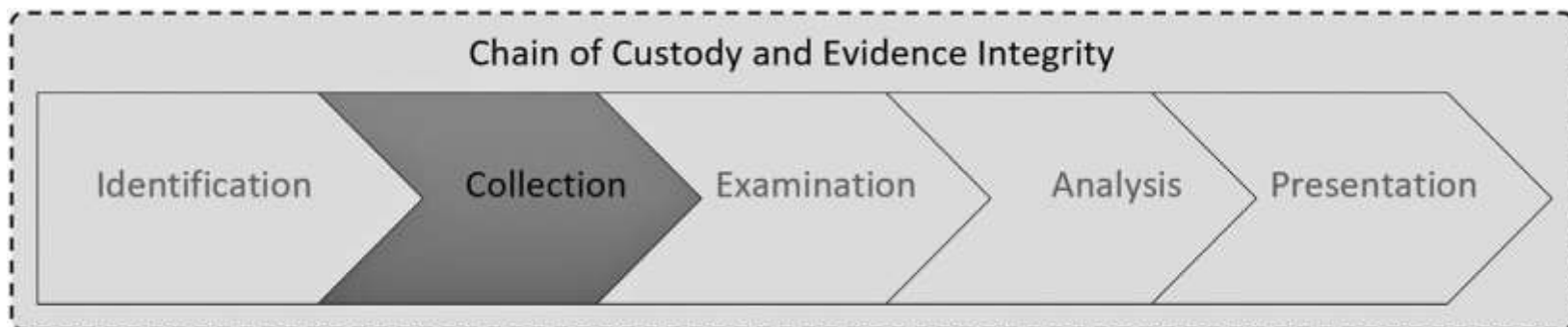
The collection phase involves acquiring or copying relevant data from electronic devices identified for the investigation.

Most digital forensics literature uses the term “**collection**,” while technical sources may refer to it as “**acquisition**” or “**extraction**.”

Copying to Separate Media: Data should always be copied to separate media for examination to prevent accidental changes to the original data.

Use of Write Blockers: Write blockers (hardware or software) are employed to ensure data is not written to the original media during the forensic process.

Digital Signatures for Integrity: Cryptographic hashes are calculated for both the original media and the copy to verify data integrity.





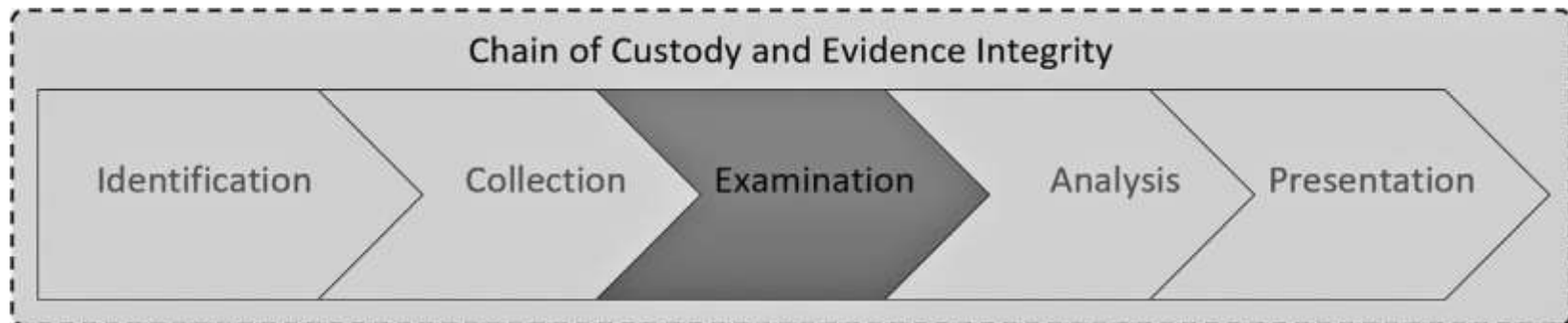
The Collection Phase...

- As errors, uncertainty, and data loss are common challenges in digital investigations, we must carefully document all.
- **Sources of Digital Evidence:** known sources of digital evidence serve as reference
- **Systems physically tied to a location:** In many cases, it is not possible to move or remove, a system or digital device from its physical location. In such cases, the data needs to be copied at the physical location itself.
- **Multiple evidence sources:** as potential digital evidence can present in different digital storage devices; we need to trace as many sources as possible.
- **Reconstruction:** as the collected data may be damaged by suspects trying to destroy evidence, or unintentionally beware that it must be reconstructed to get a complete one (partial evidence issues).
- **Evidence Integrity and Cryptographic Hashes:** evidence integrity is at the core of every forensic investigation, so to verify integrity digital fingerprinting and cryptographic algorithms apply.
- **Order of volatility:** Prioritization of the potential evidence source to be collected according to the volatility of the data.
- **Dual tool verification:** helps to detect errors from one tool by using another tool to confirm the results.



The Examination Phase

- After Preparation and extraction of potential digital evidence from collected data sources. All data collected must be examined and prepared for later analysis as **part of the examination phase**.
- It is essential to document all actions and data handling to maintain a **clear chain of custody** throughout the forensic process.
- This phase involves restructuring, parsing, and preprocessing raw data using appropriate forensic tools to retrieve relevant potential evidence, as outlined by ISO standards.



The digital forensics process: Examination



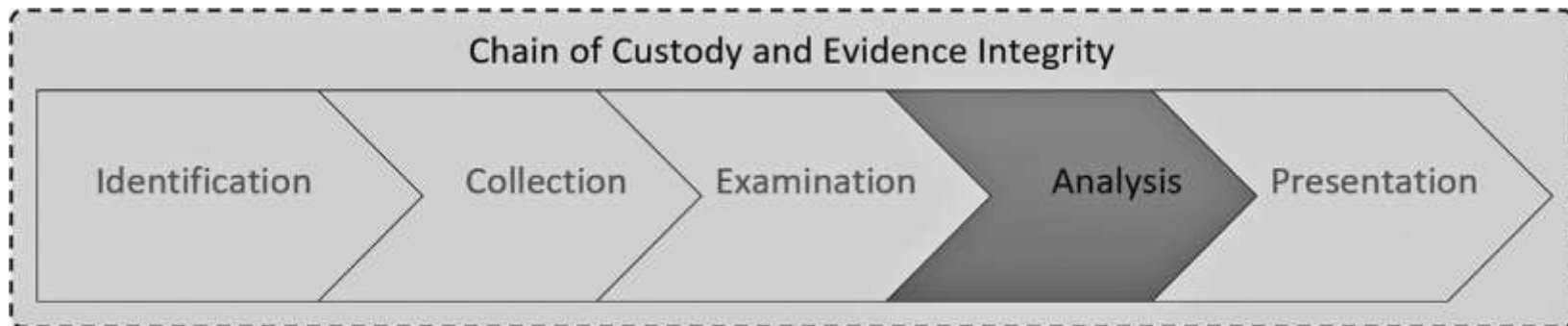
The Examination Phase...

- **Initial Data Source Examination and Preprocessing:** to reduce mistakes such as overlooking essential evidence or damaging it.
- **Forensics file formats and structures:** understand the different forensics file formats.
- **Data recovery:** if some files were erased some time ago, it is still possible to extract them as part of the investigation.
- **Data reduction and filtering:** apply data reduction using cryptographic techniques and others.
- **Time stamps:** systems internal clocks to avoid any deviation in time zones or suspicions of incorrect system clocks.
- **Compression, Encryption, and Obfuscation:** compressed files should be uncompressed, and encrypted files decrypted if possible to allow for proper analysis.
- **Data and File Carving (Diskpart):** is a data recovery technique used in digital forensics and data analysis to extract files or fragments of files from unallocated or deleted space on a storage device.



The Analysis Phase

- aims to determine the facts of an event, assess the importance of the evidence, and identify those responsible. This helps clarify what happened and who is accountable.
- examine statistical methods and manual analysis linking multiple data objects through data mining, and using timelining techniques.
- It is an **iterative phase** that **starts with preliminary hypotheses but may form new ones during the process**, leading to the collection of additional data until sufficient results are obtained for the investigation.



The digital forensics process: Analysis



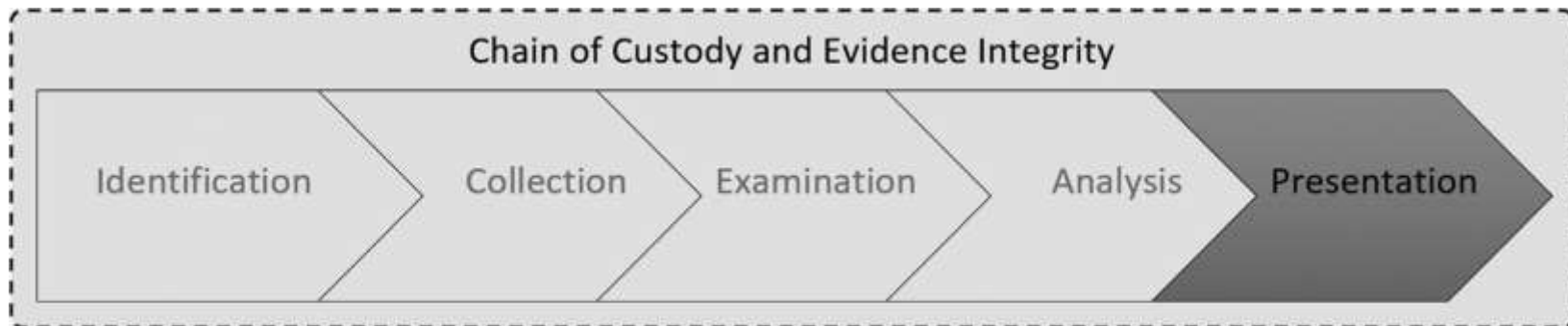
The Analysis Phase...

- **Layer of Abstraction:** to recognize that the data seen by applications, operating systems, and storage devices can differ, as files may be fragmented, deleted, or overwritten in various ways, **but** parts of these files may still be recoverable.
- **Evidence Types:** as digital evidence can be seized for you to analyze the potential evidence, beware that sources are different with different potentials.
- **Automated Analysis:** as large data volumes grow, the techniques to remove their traces make it time-consuming, costly it may be difficult for forensics to identify and analyze relevant evidence, thus we have automations.
- **Timeline of Evidence:** is a powerful tool for forensic analysis and contextual awareness.



The Presentation Phase

- Is the process by which the examiner shares results from the analysis phase in the form of reports to the interested party or parties.
- findings are documented and presented to relevant audiences, such as a court of law or corporate management teams.
- is grounded in objective evidence with a high level of certainty based on the digital evidence analyzed.
- Findings and investigative actions are summarized in a clear, understandable manner for the audience, ensuring transparency and traceability.



The digital forensics process: presentation



The Presentation Phase...

The final Reports: at this stage, the evidence is “in your hands” and is ready to put the suspect behind bars, but you should prepare the final report.

Presentation of Evidence and Work Conducted: visualization techniques are valuable in order to identify patterns and information that are not immediately obvious.

The Chain of Custody Circle Closes: The documented chain of custody is the glue that holds the forensics process together and supports the final evidence integrity so that it can be presented as trustworthy evidence in court.



Summary

Why We need a process
Principles of Digital Forensics
The Five Phases of the Process



• **Thank you very much!**